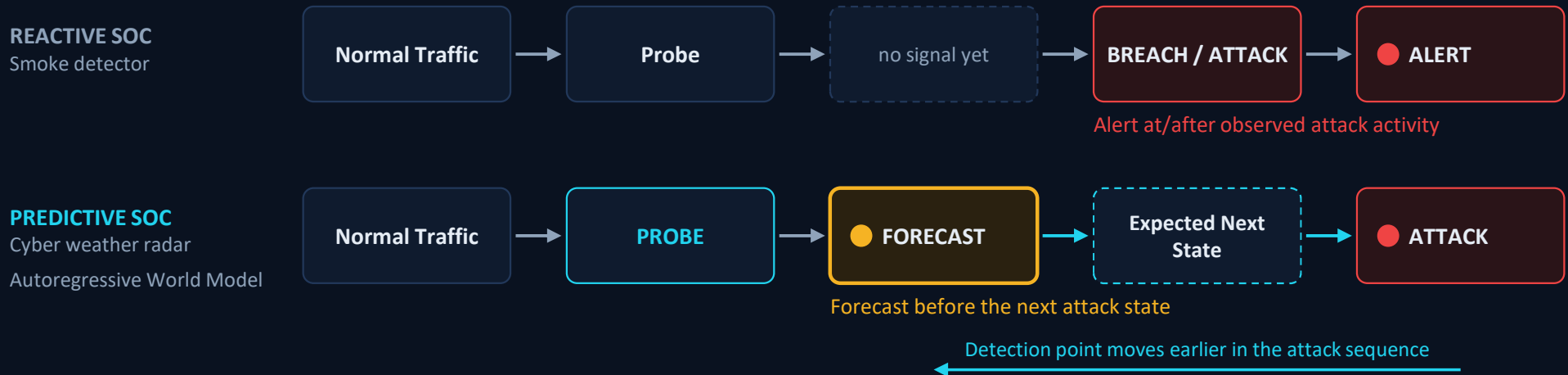


# Predictive SOC: From Reactive Alerts to Latent World Models



SMART INDIA  
HACKATHON  
2026

Team Temporal Sentinels | SIH26153 (NTRO)



SMOKE DETECTOR | reactive IDS

*"Is this traffic malicious?"*

Alerts once activity is observable

WEATHER RADAR | predictive SOC

*"Where is the network heading next?"*

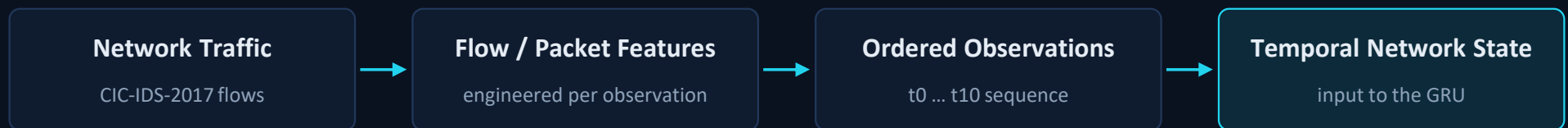
Learns temporal dynamics; forecasts K steps ahead

# Capturing the Sequence of an Attack

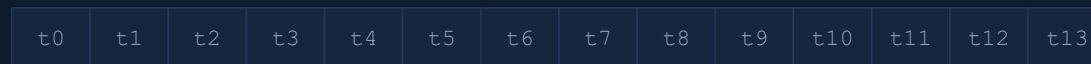
Dataset: **CIC-IDS-2017** | labeled benign traffic and multiple attack categories



SMART INDIA  
HACKATHON  
2026



## 10-second sliding windows



Window 1 | 10 s

Window 2 | 10 s

Window 3 | 10 s

*Temporal order is preserved.*

stride illustrative

## Feature families

Flow statistics

Packet-level metrics

Flag ratios

Inter-packet timing

Port behavior

Traffic volume

Connection characteristics

Feature families are illustrative examples; the final feature set follows verified extraction from CIC-IDS-2017.

# Forecasting State Transitions with GRU

PyTorch **GRU** | K-step autoregressive rollout



SMART INDIA  
HACKATHON  
2026

STATIC CLASSIFICATION

Traffic → Attack / Benign

THIS WORK | TEMPORAL STATE TRANSITIONS

$$P(S_{t+1} | S_t)$$

Current state

Predicted future states (fed back autoregressively)



K-step forecasting horizon  $\approx$  20–30 s

Horizon, not a guarantee an attack occurs within this period.

## Why GRU?

- Temporal memory
- Fewer parameters than many LSTM configurations
- Suited to low-latency sequential inference

## Detection

Is malicious activity occurring now?

## Classification

Which class is this traffic?

## Forecasting

How will the network state evolve?

## Future-state prediction

Which state is likely K steps ahead?

# Defeating the Black-Box Problem

Explainability as a trust layer between the forecasting engine and the SOC analyst



SMART INDIA  
HACKATHON  
2026



## Top driving features

*Illustrative feature attribution*

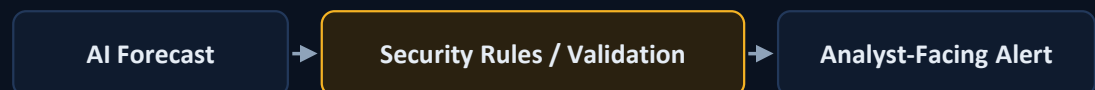


Features contributing to the model output, not proof of causality.

## MITRE ATT&CK context (conceptual progression)



Only prototype-validated mappings are shown live; CIC-IDS-2017 has no native ATT&CK labels.



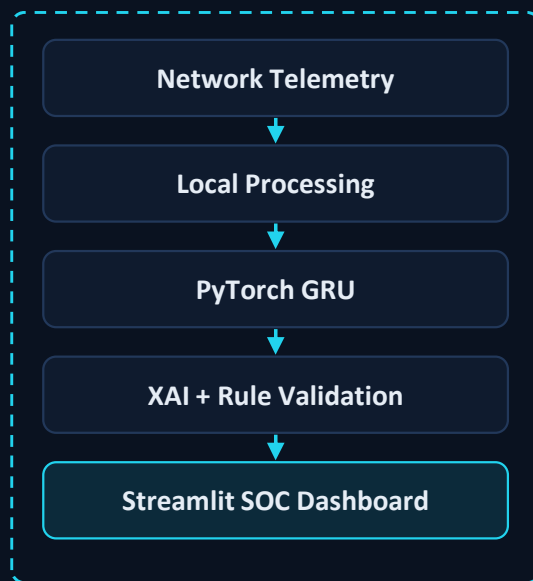
*Rule-based validation reduces unsupported AI alerts.*

# Operational Viability & Benchmarks



SMART INDIA  
HACKATHON  
2026

## Air-Gapped SOC Deployment



- 100% offline deployment
- No cloud APIs
- No external inference dependency
- Open-source software stack
- Suitable for controlled / isolated environments

## Empirical Benchmark

| Model                        | Precision | Recall   | F1       |
|------------------------------|-----------|----------|----------|
| Logistic Regression Baseline | [INSERT]  | [INSERT] | [INSERT] |
| GRU World Model              | [INSERT]  | [INSERT] | [INSERT] |

*Metrics to be populated from final experimental evaluation.*

Placeholders shown until measured values are supplied; no figures are estimated.

Designed for deployment in controlled or air-gapped environments. Security accreditation is validated separately.

**From offline telemetry → temporal modeling → future-state forecasting → explainable SOC decision support.**